task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py

```python
import os
import platform
import time
import threading

STUDENT_ID = "230103227"
INCREMENTS = 50_000_000

logical_cores = os.cpu_count() or 1
physical_cores = max(1, logical_cores // 2)
N = physical_cores

shared_counts = [0] * N

def unpadded_worker(index):  1 usage
    for _ in range(INCREMENTS):
        shared_counts[index] += 1

def local_worker(index):  1 usage
    local_count = 0
    for _ in range(INCREMENTS):
        local_count += 1
    shared_counts[index] = local_count

def run_test(worker_function):  2 usages
    threads = []
    start = time.perf_counter()

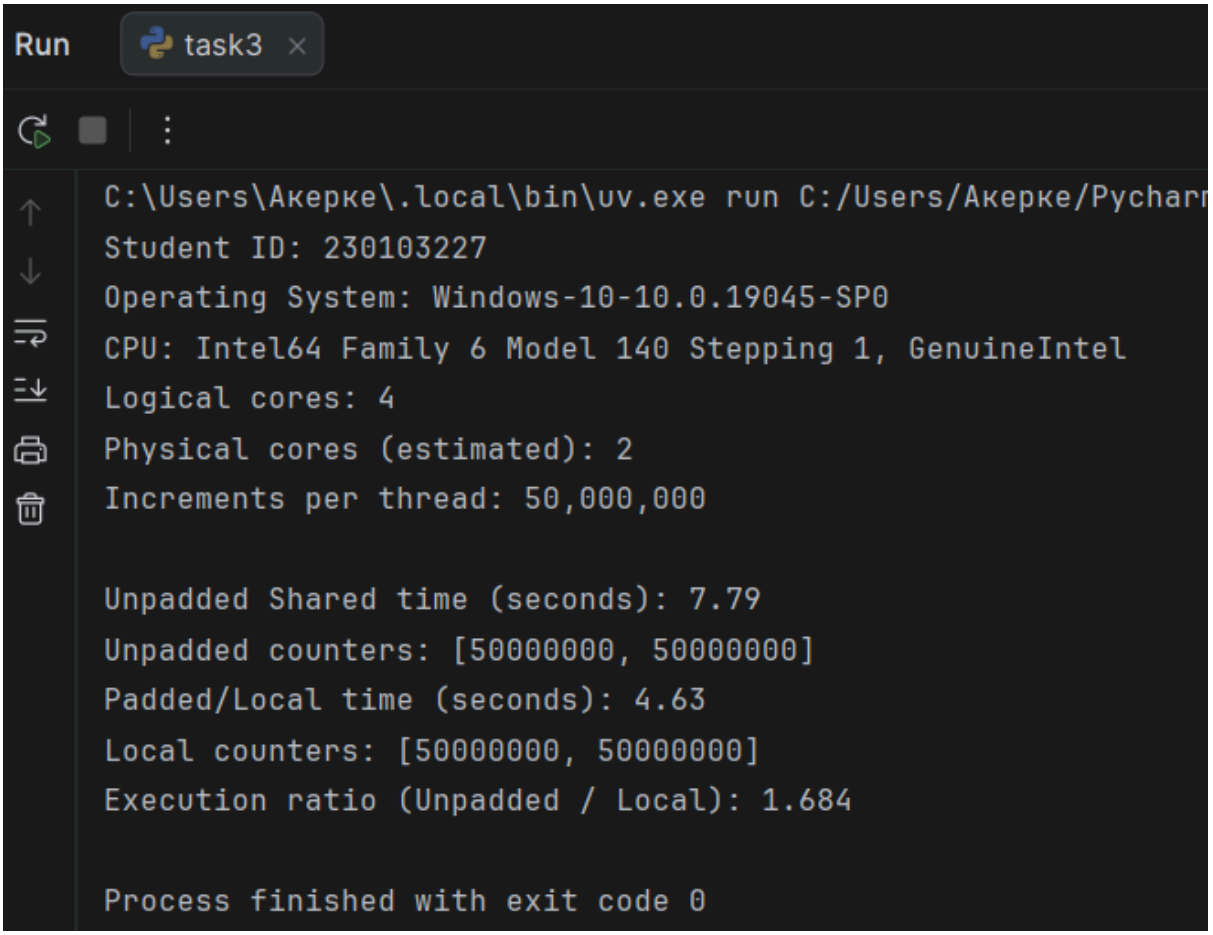
    for i in range(N):
        thread = threading.Thread(target=worker_function, args=(i,))
```

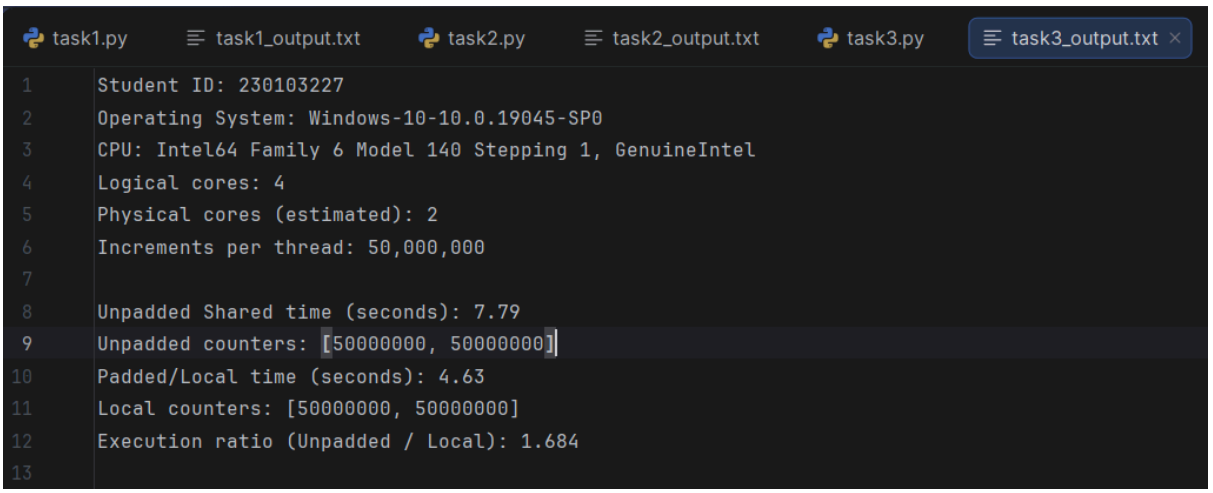
task1.py | task1_output.txt | task2.py | task2_output.txt | task3.py

```python
def run_test(worker_function):  2 usages
    for i in range(N):
        thread = threading.Thread(target=worker_function, args=(i,))
        threads.append(thread)
        thread.start()
    for thread in threads:
        thread.join()
    return time.perf_counter() - start
print("Student ID:", STUDENT_ID)
print("Operating System:", platform.platform())
print("CPU:", platform.processor() or platform.machine())
print("Logical cores:", logical_cores)
print("Physical cores (estimated):", N)
print("Increments per thread:", f"{INCREMENTS:,}")
print()
shared_counts = [0] * N
unpadded_time = run_test(unpadded_worker)
print("Unpadded Shared time (seconds):", round(unpadded_time, 2))
print("Unpadded counters:", shared_counts)
shared_counts = [0] * N
local_time = run_test(local_worker)

print("Padded/Local time (seconds):", round(local_time, 2))
print("Local counters:", shared_counts)

if local_time > 0:
    print("Execution ratio (Unpadded / Local):", round(unpadded_time / local_time, 3))
else:
    print("Execution ratio: undefined")
```


Run
task3
C:\Users\Акерке\.local\bin\uv.exe run C:/Users/Акерке/Pycharr
Student ID: 230103227
Operating System: Windows-10-10.0.19045-SP0
CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
Logical cores: 4
Physical cores (estimated): 2
Increments per thread: 50,000,000
Unpadded Shared time (seconds): 7.79
Unpadded counters: [50000000, 50000000]
Padded/Local time (seconds): 4.63
Local counters: [50000000, 50000000]
Execution ratio (Unpadded / Local): 1.684
Process finished with exit code 0


task1.py
task1_output.txt
task2.py
task2_output.txt
task3.py
task3_output.txt
1 Student ID: 230103227
2 Operating System: Windows-10-10.0.19045-SP0
3 CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
4 Logical cores: 4
5 Physical cores (estimated): 2
6 Increments per thread: 50,000,000
7
8 Unpadded Shared time (seconds): 7.79
9 Unpadded counters: [50000000, 50000000]
10 Padded/Local time (seconds): 4.63
11 Local counters: [50000000, 50000000]
12 Execution ratio (Unpadded / Local): 1.684
13